Teacher Wu Kuan | See It at a Glance #14 — Why π = 180° (Radians)

English lecture notes · YouTube Short vuqpuFxZWo4

1. The trap question

π ≈ 3.14, so why “π = 180°”?

2. The method

• 1 radian: the angle whose arc equals the radius (≈ 57.3°).
• Half circle: arc πr = π radii → π radians = 180°.
• Convert: degrees × π/180 = radians. Arc length s = rθ (θ in radians).

3. Parent check method

Ask your child to explain “π radians” as “π radius-lengths of arc”, not “3.14 degrees”.

Practice

1. Convert 90° to radians.
2. Convert π/3 to degrees.
3. Convert 270° to radians.
4. Radius 5, central angle 2 radians. Find the arc length.
5. Radius 6, central angle π/3. Find the arc length.

Answers

1. π/2
2. 60°
3. 3π/2
4. 10
5. 2π

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